```
}
}
```

Slip 19

1) Define an Interface Shape with abstract method area(). Write a java program to calculate an area of Circle and Sphere. (use final keyword).

```
interface Shape{
void area();
}
class Circle implements Shape{
final double PI=3.142;
final double radius;
Circle(double radius){
this.radius=radius;
}
public void area(){
System.out.println("Area of Circle: "+(PI*radius*radius));
}
}
class Sphere implements Shape{
final double PI=3.142;
final double radius;
Sphere(double radius){
this.radius=radius;
}
public void area(){
System.out.println("Area of Sphere: "+(4*PI*radius*radius));
}
```

```
}
public class ShapeDemo{
public static void main(String[] args){
Circle c=new Circle(5);
Sphere s=new Sphere(5);
c.area();
s.area();
}
}
```

2) Write a Java program to store student names and their Roll numbers using an appropriate collection and perform following operations: i. Add a new student and its Roll number (No duplicates) ii. Remove a student from the collection iii. Search for a student name and display the roll number.

```
import java.util.*;
public class StudentCollection{
public static void main(String[] args){
HashMap<String,Integer> students=new HashMap<>();
students.put("Rahul",101);
students.put("Priya",102);
students.put("Amit",103);
System.out.println("Students: "+students);
students.putIfAbsent("Sneha",104);
System.out.println("After adding: "+students);
students.remove("Amit");
System.out.println("After removing Amit: "+students);
String name="Priya";
if(students.containsKey(name))
System.out.println("Roll number of "+name+": "+students.get(name));
else
```

```
System.out.println("Student not found");
}
}
```

OR

2) Write a java program to demonstrate Synchronization Example.

```
class Table{
synchronized void printTable(int n){
for(int i=1;i<=5;i++){
System.out.println(n*i);
try{
Thread.sleep(400);
}catch(InterruptedException e){
System.out.println(e);
}
}
}
}
class Thread1 extends Thread{
Table t;
Thread1(Table t){
this.t=t;
}
public void run(){
t.printTable(5);
}
}
class Thread2 extends Thread{
Table t;
```